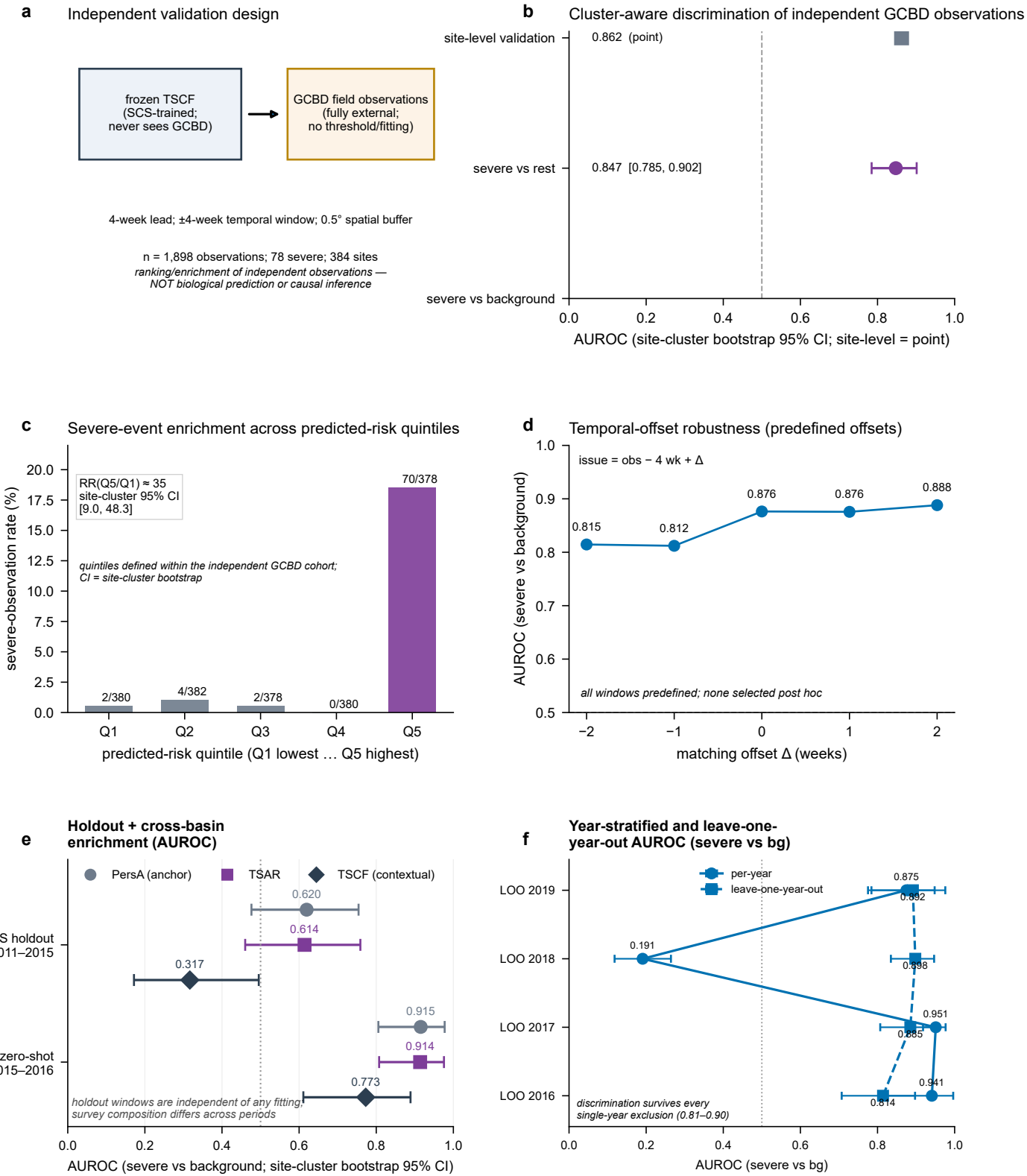

a
Independent validation design
frozen TSCF
(SCS-trained;
never sees GCBD)
GCBD field observations
(fully external;
no threshold/fitting)
4-week lead; ±4-week temporal window; 0.5° spatial buffer
n = 1,898 observations; 78 severe; 384 sites
ranking/enrichment of independent observations —
NOT biological prediction or causal inference
b
Cluster-aware discrimination of independent GCBD observations
site-level validation
0.862 (point)
severe vs rest
0.847 [0.785, 0.902]
severe vs background
0.0
0.2
0.4
0.6
0.8
1.0
AUROC (site-cluster bootstrap 95% CI; site-level = point)
c
Severe-event enrichment across predicted-risk quintiles
RR(Q5/Q1) ≈ 35
site-cluster 95% CI
[9.0, 48.3]
quintiles defined within the independent GCBD cohort;
CI = site-cluster bootstrap
severe-observation rate (%)
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
2/380
4/382
2/378
0/380
70/378
Q1
Q2
Q3
Q4
Q5
predicted-risk quintile (Q1 lowest … Q5 highest)
d
Temporal-offset robustness (predefined offsets)
issue = obs − 4 wk + Δ
AUROC (severe vs background)
1.0
0.9
0.8
0.7
0.6
0.5
0.815
0.812
0.876
0.876
0.888
all windows predefined; none selected post hoc
−2
−1
0
1
2
matching offset Δ (weeks)
e
Holdout + cross-basin
enrichment (AUROC)
PersA (anchor)
TSAR
TSCF (contextual)
SCS holdout
2011–2015
CARIB zero-shot
2015–2016
0.620
0.614
0.317
0.915
0.914
0.773
holdout windows are independent of any fitting;
survey composition differs across periods
0.0
0.2
0.4
0.6
0.8
1.0
AUROC (severe vs background; site-cluster bootstrap 95% CI)
f
Year-stratified and leave-one-
year-out AUROC (severe vs bg)
per-year
leave-one-year-out
LOO 2019
LOO 2018
LOO 2017
LOO 2016
0.875
0.892
0.191
0.898
0.951
0.885
0.941
0.614
discrimination survives every
single-year exclusion (0.81–0.90)
0.0
0.2
0.4
0.6
0.8
1.0
AUROC (severe vs bg)